Hosting DopaX on Azure

Region selection and cost at scale · list prices read from the Azure Retail Prices API on 17 August 2026

Summary

The deployment targets **Azure Container Apps** for the API, staff console and background workers, **PostgreSQL Flexible Server** as the system of record, and **Blob Storage** for the raw sensor archive, all in **Israel Central** so that participant health data stays in the country. The infrastructure is written and validated in `infra/main.bicep`; deployment is blocked only on Azure access.

$85 PER MONTH, 43 PARTICIPANTS, YEAR 1	**$1,307** PER MONTH AT 1,000 PARTICIPANTS	**3.7 TB** ARCHIVE HELD AFTER 12 MONTHS	**30–52%** SHARE OF THE BILL THAT IS STORAGE

All figures are month-12 monthly run rates. Storage accumulates indefinitely, so the monthly total rises every month the study runs.

Read the storage assumption before using any number here

The archive is the only line that grows without bound, so the assumed data volume decides most of this report — and it is unmeasured. The migration plan records that daily ZIPs *can reach* 1–2 GB, a ceiling on the heaviest days, and the legacy Drive folder has never been inventoried. These figures assume a mean of 0.4 GB per uploading participant-day at 60% adherence; the last column of the next table applies the ceiling as an average instead, which is about 2.3× higher. Reading the real folder size is the highest-value action available and needs no Azure access. Today the bill splits fairly evenly — archive $26, database $21, warm API replica $20, ingestion $16 — so storage only dominates with scale and time.

Cost by study size — Israel Central

Database tier and API replica count are derived from enrolment rather than chosen by hand, so each row is internally consistent. Every row assumes 12 months of elapsed study, the lifecycle rule in the template, and that nothing is ever deleted.

Participants	Database tier	New data / mo	Held at 12 mo	Platform / mo	Storage + egress	All-in / mo	If 1.5 GB/day
43	Burstable B1ms	314 GB	3.7 TB	$59	$26	**$85**	$156
100	Burstable B1ms	730 GB	8.5 TB	$80	$60	**$140**	$306
250	Burstable B2s	1.8 TB	21.4 TB	$191	$151	**$342**	$756
500	GP D2ds_v5	3.6 TB	42.8 TB	$462	$308	**$770**	$1,599
1,000	GP D2ds_v5	7.1 TB	85.5 TB	$681	$626	**$1,307**	$2,964
2,500	GP D4ds_v5	17.8 TB	213.8 TB	$1,499	$1,577	**$3,076**	$7,204
5,000	GP D8ds_v5	35.6 TB	427.5 TB	$3,366	$3,163	**$6,529**	$14,745
10,000	GP D8ds_v5	71.2 TB	855.0 TB	$5,205	$6,316	**$11,521**	$27,930

Participants	Database tier	New data / mo	Held at 12 mo	Platform / mo	Storage + egress	All-in / mo	If 1.5 GB/day
20,000	GP D16ds_v5	142.5 TB	1.67 PB	$10,242	$12,600	**$22,842**	$55,631

Growth uses 0.4 GB per uploading participant-day at 60% adherence, so one participant produces about 7 GB a month. The last column is the same arithmetic with the 1–2 GB ceiling from `backend/docs/MIGRATION_PLAN.md` §2.3 used as a daily average, which is what the first version of this report did. Rows at 10,000 and above are arithmetic for completeness, not a plan; see the ceiling below.

Above roughly 2,500 participants this architecture is the wrong answer

Three things break well before the 20,000 row. Ingest reaches 142.5 TB a month, roughly 4.7 TB a day arriving — a sustained feed rather than background uploads from phones. Ingestion becomes 364,800 job executions a month, past what Container Apps Jobs is designed to schedule; the design would move to a queue with a long-lived consumer pool. And nobody buys storage at this volume at list price: Azure reserved capacity at 100 TB or 1 PB commitments discounts 25–38%, which these figures deliberately exclude. Most importantly, retaining every raw sensor ZIP forever is affordable at 43 participants and indefensible at 20,000, where it means 1.67 PB after one year. Long before that scale the study would keep derived features and discard raw signal after a window — a research-design decision, not an infrastructure one.

Which region is cheapest

All-in monthly spend in month 12 at 1,000 participants, each region using the coldest storage tier it actually offers. Israel Central is the only option that satisfies in-country residency, and it is also the most expensive on almost every meter.

Region	Residency	Coldest tier	Platform / mo	Storage / mo	All-in / mo	vs cheapest
Sweden Central	Outside Israel	Archive $0.00099	$519	$345	**$887**	—
North Europe	Outside Israel	Archive $0.00099	$524	$371	**$918**	+$31
Germany West Central	Outside Israel	Archive $0.0018	$537	$407	**$967**	+$80
East US	Outside Israel	Archive $0.00099	$506	$439	**$968**	+$81
France Central	Outside Israel	Archive $0.00211	$531	$430	**$985**	+$98
West Europe	Outside Israel	Archive $0.0018	$671	$407	**$1,101**	+$214
UK South	Outside Israel	Archive $0.0018	$665	$417	**$1,106**	+$219
Israel Central	In Israel	Cold $0.0045 — no Archive	$681	$603	**$1,307**	+$420

Shaded green: cheapest. Shaded amber: the in-country option, which is the one the template uses.

What residency costs, by study size

Participants	Israel Central	Sweden Central	Premium	%
43	$85	$63	**+$22**	+35%
250	$342	$210	**+$132**	+63%
1,000	$1,307	$887	**+$420**	+47%
5,000	$6,529	$4,405	**+$2,124**	+48%
20,000	$22,842	$15,017	**+$7,825**	+52%

The premium runs 35–63% across the range, so it will not improve with scale.

Why Israel Central costs more

Three structural differences. Container Apps compute is priced 42% above the cheap European tier — $3.4e-05 per vCPU-second against $2.4e-05 — and because ingestion runs one job per uploaded participant-day, that premium tracks enrolment directly. The database is 11% more per hour for the same burstable SKU. Most significantly, **Israel Central offers no Archive blob tier at all**, so Cold LRS at $0.0045/GB-month is the floor, against $0.00099 in Sweden or North Europe — 4.5× on the tier that eventually holds nearly all of the data. Logging is no longer a factor now that console logs sit on the Basic plan: $0.66/GB against $0.645.

> **The one lever that would change the storage bill**
> If the ethics approval permits de-identified raw ZIPs to sit in the EU while identifiable records stay in PostgreSQL in Israel, a split deployment becomes worth costing: database and API in Israel Central, cold archive in Sweden Central or North Europe on Archive LRS at $0.00099 — about a 43% cut to the storage line at 1,000 participants. This is a question for the ethics board, not a technical decision. The template deliberately keeps everything in Israel Central, which is the conservative and defensible default. Moving participant data across a border on cost grounds alone is exactly the kind of change that fails an audit.

What gets provisioned today — 43 participants, Israel Central

Component	Service and SKU	Unit price	Est. / month
Backend API	Container Apps · 0.5 vCPU / 1 GiB, 1 always warm	$3.4e-05/vCPU-s active, $4e-06 idle; memory billed all month	$20
Admin console	Container Apps · 0.5 vCPU / 1 GiB, scales to zero	Billed only while staff are using it	$3
Ingestion jobs	Container Apps Jobs · 784 runs/mo, 4 min at 2 vCPU	60% of participant-days upload	$16
Reconciler job	Container Apps Jobs · nightly cron, 10 min at 1 vCPU	Guardrail R3 counts 14 clean runs	$1
Free grant	180k vCPU-s + 360k GiB-s per subscription	Applied once across all apps and jobs	–$8
Database	PostgreSQL Flexible Server · Burstable B1ms	$0.0220/hr + $0.164/GB-mo × 32 GB provisioned	$21
Raw ZIP archive	Blob Storage GPv2 LRS · 3.7 TB at month 12	Hot $0.0184–0.02 tiered · Cool $0.01104 · Cold $0.0045	$26
Image registry	Container Registry · Basic	$0.1666/day	$5
Secrets	Key Vault Standard	$0.0396 per 10k operations	$1
Logs	Log Analytics Basic plan · 0.5 GB/mo, capped at 0.5 GB/day	$0.66/GB Basic instead of $3.29/GB Analytics	$0
Researcher downloads	Egress · 16 GB/mo at 5% of new data	First 100 GB free, then $0.087/GB	$0
Participant uploads	Inbound data transfer	Ingress is always free, at any volume	$0

Every line is a real Azure meter at pay-as-you-go list price. Storage is shown at its month-12 volume.

What this model assumes

1. **The data volume is unmeasured, and it decides the report.** `MIGRATION_PLAN.md` §2.3 records that daily ZIPs *can reach* 1–2 GB. That is a ceiling on the heaviest days, not an average, and the legacy Drive folder has never been inventoried. These figures assume a mean of 0.4 GB per uploading participant-day and 60% adherence, both estimates. Reading the real folder size and file count replaces the largest source of error in this document and requires no Azure access at all.
2. **Nothing is ever deleted.** Matching the lifecycle policy in the template, which tiers but never expires. The retention policy for raw ZIPs is still an open question in the migration plan (§13). Retiring raw archives after twelve months turns the storage line from a permanently rising cost into a flat one, and is the second-largest lever after the volume assumption.
3. **Hot storage is volume-tiered; Cool and Cold are not.** Hot bills $0.02/GB-month up to 50 TB, $0.0192 to 500 TB, then $0.0184 in Israel Central, applied progressively. Cool, Cold and Archive return a single flat tier from the pricing API.
4. **Ingestion is the biggest guess in the compute model.** One execution per uploaded participant-day, four minutes at 2 vCPU and 4 GiB. At 43 participants that is $16 a month; at 2,500 it is a top-three line and worth measuring against a real ZIP before anyone budgets on it.
5. **Egress is an estimate, and it was missing entirely from the first version of this report.** It assumes researchers pull back 5% of each month's new data for analysis, against a 100 GB monthly free allowance, at the most expensive band. Uploads into Azure are free at any volume, so data collection itself costs nothing to receive.
6. **Compute is sized for the load, not for reassurance.** The database is Burstable B1ms up to 150 participants because raw ZIPs live in blob storage and the server holds only metadata — roughly 30 rows per participant-day. One API replica is kept warm deliberately: Container Apps bills allocated capacity per second, so a warm replica costs about $20 a month whether

traffic arrives or not, and scaling it to zero would save roughly $14 at the risk of a cold start in front of a phone uploading on cellular. The staff console does scale to zero.

7. **Sweden's Burstable rate looks anomalous.** B2s prices at $0.0398/hr against $0.088 in Israel — less than half, while its General Purpose rate is only 14% cheaper. Worth verifying in the portal before relying on the regional comparison at small enrolments.
8. **Quota is not the same as availability.** Every service and SKU here was confirmed present in Israel Central via the pricing API, but a new subscription commonly starts with zero cores in the region. Container Apps vCPU and Burstable vCores both need confirming before the first deploy.
9. **List prices only.** No EA or CSP discount, no reserved instances, no savings plan. A one-year reservation cuts General Purpose PostgreSQL compute by roughly 40%, and storage reserved capacity discounts 25–38%. Both matter from the 250-participant row upward and neither is included.
10. **The HIPAA BAA covers Azure, not Firebase.** Container Apps, PostgreSQL, Blob Storage, Key Vault and Container Registry are all in scope at no extra cost. Firebase Auth remains the identity provider under guardrail R1, so participant emails stay with Google and outside that BAA — a decision to record deliberately rather than discover during an audit.

Deployment status

Piece	State
`backend/Dockerfile`	Written. Multi-stage Node 22; runtime, release and worker targets; migrations copied in.
`admin/Dockerfile`	Written. Next.js standalone output, builds and runs.
`infra/main.bicep`	Written, compiles clean. Two-pass deploy via `deployApplications`. Right-sized: B1ms database on a 32 GB floor, console logs on the Basic plan, staff console scaling to zero.
`infra/deploy.sh`	Written. Idempotent; supports `--what-if` and `--infra-only`.
`.github/workflows/deploy.yml`	Written. Lint, typecheck, test, build, push, deploy on federated credentials.
Azure subscription	**Missing.** The tenant has no subscription, so nothing can be deployed yet. This is the only blocker.
`backend/src/storage/`	Absent. `env.ts` validates `STORAGE_BACKEND=azure` but nothing implements it.
Reconciler entrypoint	Placeholder. The job is scheduled so the cron is reviewable; the command is a stub.

What is needed from the Azure administrator

1. A **subscription**, and its subscription ID. If an Enterprise Agreement or Microsoft Customer Agreement already exists, this needs no new payment method.
2. **Owner on a dedicated resource group**, not Contributor. The template creates three role assignments to bind the workload identity to the registry, storage account and vault; that requires `Microsoft.Authorization/roleAssignments/write`, which Contributor excludes. With Contributor the deployment fails midway, after the database exists.
3. **Resource providers registered** at subscription scope: `Microsoft.App`, `ContainerRegistry`, `DBforPostgreSQL`, `Storage`, `KeyVault`, `ManagedIdentity`, `OperationalInsights`, `Insights`, `Network`.
4. **Israel Central permitted and quota confirmed** — at least 10 vCPU for Container Apps and 2 Burstable vCores for PostgreSQL. Quota requests take a day or two.
5. **Confirmation that no policy forbids public endpoints** on managed services. If private endpoints are mandated, that is a design change (VNet integration) and needs deciding before the build, not after.
6. **Confirmation that the HIPAA BAA covers the subscription.**
7. **Budget approval and a budget alert** on the resource group.

The full request, with the exact commands, is in `infra/AZURE_ACCESS_REQUEST.md` (Hebrew version: `AZURE_ACCESS_REQUEST.he.md`).

All unit prices are pay-as-you-go list rates read directly from the Azure Retail Prices API on 17 August 2026, filtered to the exact meters consumed by the resources declared in `infra/main.bicep`, across the eight regions compared. They exclude EA and CSP discounts, reserved instances and savings plans. Compute sizing is an estimate for a low-traffic internal study, not a quote. Storage growth assumes 0.4 GB per uploading participant-day at 60% adherence — an estimate, not a measurement — and that nothing is ever deleted, matching the lifecycle policy in the template. Re-check any unit price with `scratch/azure_unit_prices.py`, which queries the pricing API directly and needs no credentials. Generated from `scratch/azure_cost_report.py`; the interactive version is the azure-hosting-plan canvas.